

Universal Feature Selection with Noisy Observations and Weak Symmetry Conditions

Dier Tang, Guangyue Han

Department of Mathematics, The University of Hong Kong

1. Introduction

- Feature selection [1] is essential in high-dimensional learning and inference.
- While task-specific features can be obtained in many ways, a more ambitious goal is to extract **universally effective** features that are applicable across a diverse range of tasks.
- Consider Markov chain $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$, we extract features based on observed variables X, Y to infer the latent attributes U, V .

2. Preliminaries

- ϵ -neighborhood of $P_Z \in \mathcal{P}^Z$:

$$\mathcal{N}_Z^\epsilon(P_Z) = \{P \in \mathcal{P}^Z : \mathcal{X}^2(P \| P_Z) \leq \epsilon^2\}.$$
- ϵ -attribute: W is an ϵ -attribute of Z , if $\forall w \in \mathcal{W}$, $P_{Z|W}(\cdot | w) \in \mathcal{N}_Z^\epsilon(P_Z)$. Its configuration

$$\mathcal{C}_Z^\epsilon(P_Z) = \{\mathcal{W}, \{P_W(w)\}, \{P_{Z|W}(\cdot | w)\}\}.$$
- Information matrix & vector: For each z_i, w_j ,

$$\Phi_{i,j}^{Z|W} = \frac{1}{\epsilon} \cdot \frac{P_{Z|W}(z_i | w_j) - P_Z(z_i)}{\sqrt{P_Z(z_i)}}.$$

Column $\phi_{w_j}^{Z|W}$ is the information vector for w_j .

- Feature vectors: Given normalized features $h^k : \mathcal{Z} \rightarrow \mathbb{R}^k$, define $\psi_i^Z(z) = \sqrt{P_Z(z)} h_i(z)$.
- Canonical dependence matrix: For X, Y , let $K_{XY} = \min\{|\mathcal{X}|, |\mathcal{Y}|\}$ and

$$(\tilde{B}_{X,Y})_{j,i} = \frac{P_{X,Y}(x_i, y_j) - P_X(x_i)P_Y(y_j)}{\sqrt{P_X(x_i)}\sqrt{P_Y(y_j)}}.$$

3. Problem Setup

Consider the generalized model,

$$\begin{array}{ccccc} U & \xleftrightarrow{\epsilon} & X & \longleftrightarrow & Y & \xleftrightarrow{\epsilon} & V \\ & & \downarrow & & \downarrow & & \\ & & \hat{X} & & \hat{Y} & & \end{array} \quad (1)$$

- U, V are the ϵ -attributes of X, Y , as an inspiration from **Information Bottleneck** [2], i.e.,

$$\max I(U; Y) \quad \text{s.t. } I(U; X) \leq \epsilon,$$
since $D_{\text{KL}}(p \| q) \leq \mathcal{X}^2(p \| q), \forall p, q$.
- The collection of $\mathcal{C}_X^\epsilon(P_X), \mathcal{C}_Y^\epsilon(P_Y)$ are equipped with probability measures μ_U, μ_V .
- X, Y is unavailable, providing only their noisy counterparts $\hat{X}, \hat{Y}$ over the same alphabets $\mathcal{X}, \mathcal{Y}$, with noisy channels

$$P_{\hat{X}|X} = \mathbf{I} + \eta_1 \mathbf{T}_{\hat{X}|X}, \quad P_{\hat{Y}|Y} = \mathbf{I} + \eta_2 \mathbf{T}_{\hat{Y}|Y} \quad (2)$$

for some $\eta_1, \eta_2 \geq 0$.

- Data $\hat{x}^N = \{\hat{x}_1, \dots, \hat{x}_N\}, \hat{y}^N = \{\hat{y}_1, \dots, \hat{y}_N\}$ are i.i.d. drawn from $P_{\hat{X}, \hat{Y}}$. Statistics $\hat{S}^k, \hat{T}^k$ are the empirical expectations of $f^k(\hat{X}), g^k(\hat{Y})$.
- Denote the error probability for inferring U from S^k by $p_e^{U|S}(f^k, \mathcal{C}_X^\epsilon(P_X))$, and the Chernoff error exponent

$$\bar{E}^{U|S}(f^k) = \lim_{N \rightarrow \infty} -\frac{\mathbb{E}_{\mu_U}[\log P_e^{U|S}(f^k, \mathcal{C}_X^\epsilon(P_X))]}{N}.$$

Similarly define $\bar{E}^{V|S}(f^k), \bar{E}^{U|T}(g^k), \bar{E}^{V|T}(g^k)$.

- We aim to seek features f^k and g^k that maximize the above exponents.

Note that when setting $\eta_1 = \eta_2 = 0$, the generalized model (1) boils down to

$$U \xleftrightarrow{\epsilon} X (\hat{X}) \longleftrightarrow Y (\hat{Y}) \xleftrightarrow{\epsilon} V. \quad (3)$$

which is the model in [3] and [4].

4. Previous Works

Recent works [3] and [4] operate within (3). They propose that it's natural to assume μ_U and μ_V exhibit **no directional preference**, i.e. $\Phi^{X|U}$ and $\Phi^{Y|V}$ satisfy spherical symmetry [5].

Definition 1 (Spherical symmetry) A random matrix $A \in \mathbb{R}^{n \times m}$ is spherically symmetric, if for any deterministic orthogonal matrices $Q_1 \in \mathbb{R}^{n \times n}$ and $Q_2 \in \mathbb{R}^{m \times m}$, we have $Q_1^T A Q_2 \stackrel{d}{=} A$.

The result of [4, Prop 5.10] is reformulated below in our notation.

Theorem 1 (Huang et al. 2024) Consider the Markov chain (3). For $k \in \{1, \dots, K_{XY}-1\}$, select normalized features $f^k : \mathcal{X} \rightarrow \mathbb{R}^k$ and $g^k : \mathcal{Y} \rightarrow \mathbb{R}^k$. Assume under each probability measures, $\Phi^{X|U}$ and $\Phi^{Y|V}$ are spherically symmetric. Then,

$$\begin{aligned} &(\bar{E}^{U|S}(f^k), \bar{E}^{V|S}(f^k), \bar{E}^{U|T}(g^k), \bar{E}^{V|T}(g^k)) \leq \\ &(C_U \epsilon^2 k, C_V \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_U \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_V \epsilon^2 k) + o(\epsilon^2), \end{aligned} \quad (4)$$

as $\epsilon \rightarrow 0$, where C_U, C_V are positive constants, and σ_i is the i -th largest singular value of $\tilde{B}_{\hat{X}, \hat{Y}}$. Moreover, the equality holds when the feature vectors corresponding to f_i and g_i are precisely the right and left singular vectors of $\tilde{B}_{\hat{X}, \hat{Y}}$ corresponding to σ_i .

5. Motivation and Observation

The exact spherical symmetry condition is rather restrictive:

Lemma 2 If A is spherically symmetric, then: (1) $\mathbb{E}[A] = 0$; (2) $\forall i, j, k, l, A_{i,j} \stackrel{d}{=} A_{k,l}$; (3) For $(i, j) \neq (k, l)$, $\text{Cov}(A_{i,j}, A_{k,l}) = 0$.

Assume W is an ϵ -attribute of Z . Expand the log-likelihood ratio

$$\log \frac{P_{Z|W}(z | w)}{P_Z(z)} = \epsilon \frac{\phi_w(z)}{\sqrt{P_Z(z)}} - \frac{\epsilon^2}{2} \left(\frac{\phi_w(z)}{\sqrt{P_Z(z)}} \right)^2 + o(\epsilon^2).$$

For any smooth family $\{w(\theta)\}$, the score function

$$\left. \frac{\partial}{\partial \theta} \log P(z | w(\theta)) \right|_{\theta=0} = \epsilon \frac{\phi'_w(z)}{\sqrt{P_Z(z)}} + O(\epsilon^2),$$

where ϕ'_w denotes the directional derivative about θ . Then, the Fisher information $I(0) = \epsilon^2 \|\phi'_w\|^2 + O(\epsilon^3)$. According to the **Cramér-Rao lower bound**, as $\epsilon \rightarrow 0$, the term $\|\phi'_w\|^2$ dominates the MSE of estimation, indicating that in the local regime, imposing spherical symmetry can be reduced to a **second-moment symmetry condition**. A weakened condition is then proposed:

Definition 2 (Weak spherical symmetry) For random matrices $X, Y \in \mathbb{R}^{n \times m}$, define a second-moment distance as

$$D(X, Y) = \sup_{u \in \mathbb{R}^n, v \in \mathbb{R}^m, \|u\|=\|v\|=1} \left| \mathbb{E}[(u^T X v)^2] - \mathbb{E}[(u^T Y v)^2] \right|.$$

Given $\delta \geq 0$, a random matrix $A \in \mathbb{R}^{n \times m}$ is δ -spherically symmetric, if for any orthogonal matrices $Q_1 \in \mathbb{R}^{n \times n}$, $Q_2 \in \mathbb{R}^{m \times m}$, we have $D(Q_1^T A Q_2, A) \leq \delta$.

6. Main Result

Theorem 3 (Tang, Han, 2026) Consider in (1), $\hat{X}, \hat{Y}$ are generated via (2) characterized by η_1, η_2 . For $k \in \{1, \dots, K_{XY}-1\}$, select normalized features $f^k : \mathcal{X} \rightarrow \mathbb{R}^k, g^k : \mathcal{Y} \rightarrow \mathbb{R}^k$. Given $\delta \geq 0$, assume under each probability measures, $\Phi^{X|U}$ and $\Phi^{Y|V}$ are δ -spherically symmetric. Then,

$$\begin{aligned} &(\bar{E}^{U|S}(f^k), \bar{E}^{V|S}(f^k), \bar{E}^{U|T}(g^k), \bar{E}^{V|T}(g^k)) \leq \\ &(C_U \epsilon^2 k, C_V \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_U \epsilon^2 \sum_{i=1}^k \sigma_i^2, C_V \epsilon^2 k) + R(\epsilon, \delta, \eta_1, \eta_2), \end{aligned} \quad (5)$$

as $\epsilon \rightarrow 0$, where

$$R(\epsilon, \delta, \eta_1, \eta_2) = O(\epsilon^2 \cdot \max\{\delta + \eta_1 + \delta\eta_1, \delta + \eta_2 + \delta\eta_2\}),$$

and C_U, C_V are positive constants, σ_i is the i -th largest singular value of $\tilde{B}_{\hat{X}, \hat{Y}}$. Moreover, the equality holds when the feature vectors corresponding to f_i and g_i are precisely the right and left singular vectors of $\tilde{B}_{\hat{X}, \hat{Y}}$ corresponding to σ_i .

7. Comments and Examples

Exact spherical symmetry implies δ -spherical symmetry. The latter is a practical relaxation:

Example 1 Given $0 < |\sigma^2 - 1| \leq \delta$. A random matrix $A \in \mathbb{R}^{2 \times 2}$ has independent entries $A_{1,1} \sim N(0, \sigma^2)$ and $A_{1,2}, A_{2,1}, A_{2,2} \sim N(0, 1)$. Then, A is δ -spherically symmetric, but not exactly spherically symmetric.

Compared to Theorem 1, the weakened condition in Theorem 3 allows the structures to exhibit bounded directional preference ($\leq \delta$).

Remark 1 If $\delta, \eta_1, \eta_2 = o(1)$ as $\epsilon \rightarrow 0$, we have $R(\epsilon, \delta, \eta_1, \eta_2) = o(\epsilon^2)$. Then, (5) reduces to (4). Specifically, when $\eta_1 = \eta_2 = 0$, (1) boils down to (3). Thus, **Theorem 3 recovers Theorem 1**, which indicates the **exact symmetry condition is unnecessary**, and a relatively small directional preference $\delta = o(1)$ is tolerable. It further implies the second-moment symmetry condition is sufficient for the original result.

8. Related Results

Lemmas 4 and 5 reveal valuable properties of weak spherical symmetry, and is useful for the proof of Theorem 3 and related analysis.

Lemma 4 Given $\delta \geq 0$, assume $A \in \mathbb{R}^{n \times m}$ is δ -spherically symmetric. For any deterministic matrices $G \in \mathbb{R}^{n \times k_1}, H \in \mathbb{R}^{m \times k_2}$, there exists a constant C independent of δ such that

$$\left| \mathbb{E}[\|G^T A H\|_F^2] - \frac{1}{mn} \|G\|_F^2 \|H\|_F^2 \mathbb{E}[\|A\|_F^2] \right| \leq C\delta.$$

Lemma 5 Given $\delta \geq 0$, assume $A \in \mathbb{R}^{n \times m}$ is δ -spherically symmetric, and $B \in \mathbb{R}^{n \times n}$ is a deterministic matrix with singular values $\sigma_1 \geq \dots \geq \sigma_n \geq 0$. Then, BA is γ -spherically symmetric, where $\gamma = (\alpha + \delta)(\sigma_1^2 - \sigma_n^2) + \sigma_1^2 \delta$ with $\alpha = \frac{1}{mn} \mathbb{E}[\|A\|_F^2]$.

Remark 2 Suppose $B = \mathbf{I}$, then $BA \stackrel{d}{=} A$, and by Lemma 5, we obtain $\gamma = \delta$. Assume A is exactly spherically symmetric, i.e., $\delta = 0$, Lemma 5 gives $\gamma = (\sigma_1^2 - \sigma_n^2)\alpha$, indicating the symmetry deviation of BA arises solely from the range among the singular values of B .

9. Conclusions

- We extend the original universal feature selection framework to more general scenarios (allow directional preference and observation noise), and recover the original result as a special case when $\eta_1 = \eta_2 = 0$ and $\delta = o(1)$.
- The selection framework is **robust** against symmetry deviation δ , and noise levels η_1, η_2 .
- Since $R(\epsilon, \delta, \eta_1, \eta_2)$ depends explicitly on δ, η_1, η_2 , the error bound is well-controlled even if they are not vanishingly small (e.g., $O(1)$).
- We believe our work provides a practical and general theory for universal feature selection in machine learning and statistical inference.

Details

Details are available at **arXiv: 2605.09396** [6].

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More Information



Main Author

Dier Tang
Department of Mathematics
The University of Hong Kong
tangde_math@connect.hku.hk
Website: <https://tangdier03.github.io>